

Builder Basics · One Variable, Any Size — English Worksheet

Topic: Building Structures From One Input · **Course:** Builder Basics · **Time:** about 45 minutes

One number at the top of your code decides the whole build. Change that one number, run again, and the same code builds a different size.

1 · Predict

Read each set of steps. Write what you think will happen.

```
size = 4
place on move ON
repeat 4 times:
    move forward by size
    turn left
```

What shape does the agent draw? How long is each side?

Now the first line says `size = 7` instead. How many lines did you edit, and how long is each side now?


```

size = 5
place on move ON
repeat size times:
    repeat 4 times:
        move forward by size
        turn left
    place on move OFF
    move up by 1
    place on move ON

```

How wide is the build? How tall is it? Which line decides both?


```

size = 8
repeat 4 times:
    [draw one square, size blocks per side]
    move up by 1
    size = size - 2

```

Write the side length of every layer, bottom to top.

Change the first line to `size = 10` . Write the new layers, bottom to top. What happens to the number of layers?

2 · Spot the Bug 🐛

Each code block is broken. Read what it should do, rewrite it, then explain why the original was wrong.

Bug A — Every side of the square should follow `size`. Right now one side ignores it.

```
size = 9
place on move ON
move forward by size
turn left
move forward by size
turn left
move forward by 6
turn left
move forward by size
turn left
```

Write the fixed code:

What shape do you get instead of a square? Why does your fix work?

Bug B — The tower should be exactly as tall as the wall beside it. The wall is 10 blocks tall.

```

size = 11
place on move ON
repeat size times:
    repeat 4 times:
        move forward by 5
        turn left
    place on move OFF
    move up by 1
    place on move ON

```

Count the layers this code builds. Write the number, then write the fixed first line.

How can you check the height before you run the code, instead of after?

Bug C — One input should change the whole house. Here, changing `size` only changes the floor.

```

size = 6
place on move ON
repeat 4 times:
    move forward by size
    turn left
place on move OFF
move up by 1
place on move ON
repeat 4 times:
    move forward by 4
    turn left

```

Write the fixed code:

Why was it wrong? Why does your fix work?

Run it three times. Between runs, edit **only** the first line: a small size, a medium size, a big size. Leave every other line alone.

1 · Your code. Scroll through it. Say what each part does. Point at the one line you edit between runs.

Fill the blanks:

I built it using this code: _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

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